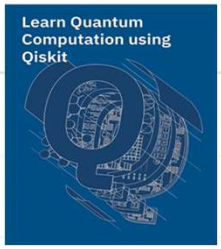
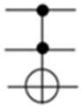
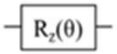
Assignment Guide

Dr. -Ing. Somrak Petchartee

Single-Qubit Gates

Gate	Symbol	Matrix	Qiskit Command
Pauli-X	X	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	`qc.x(q)`
Pauli-Y	Y	$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$	`qc.y(q)`
Pauli-Z	Z	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	`qc.z(q)`
Hadamard	H	$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$	`qc.h(q)`
S	S	$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$	`qc.s(q)`
S†	S†	$\begin{bmatrix} 1 & 0 \\ 0 & -i \end{bmatrix}$	`qc.sdg(q)`
T	T	$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$	`qc.t(q)`
T†	T†	$\begin{bmatrix} 1 & 0 \\ 0 & e^{-i\pi/4} \end{bmatrix}$	`qc.tdg(q)`
Rx(θ)	$R_x(\theta)$	$\begin{bmatrix} \cos\frac{\theta}{2} & -i\sin\frac{\theta}{2} \\ -i\sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix}$	`qc.rx(theta,q)`
Ry(θ)	$R_y(\theta)$	$\begin{bmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix}$	`qc.ry(theta,q)`
Rz(θ)	$R_z(\theta)$	$\begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix}$	`qc.rz(theta,q)`



Gate name	# Qubits	Circuit Symbol	Unitary Matrix	Description
Hadamard	1		$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$	Transforms a basis state into an even superposition of the two basis states.
T	1		$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$	Adds a relative phase shift of π/4 between contributing basis states. Sometimes called a π/8 gate, because diagonal elements can be written as $e^{-i\pi/8}$ and $e^{i\pi/8}$.
CNOT	2		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	Controlled-not; reversible analogue to classical XOR gate. The input connected to the solid dot is passed through to make the operation reversible.
Toffoli (CCNOT)	3		$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$	Controlled-controlled-not; a three-qubit gate that switches the third bit for states where the first two bits are 1 (that is, switches |110⟩ to |111⟩ and vice versa).
Pauli-Z	1		$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	Adds a relative phase shift of π between contributing basis states. Maps |0⟩ to itself and |1⟩ to −|1⟩. Sometimes called a "phase flip."
Z-Rotation	1		$\begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix}$	Adds a relative phase shift of (or rotates state vector about z-axis by) θ.
NOT	1		$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	Analogous to classical NOT gate; switches |0⟩ to |1⟩ and vice versa.

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Basics

3 qubits → statevector has $2^3 = 8$ amplitudes, one per basis state, ordering: $|q_2\, q_1\, q_0\rangle$, so:

Index	Binary	Meaning
0	000	$q_2=0, q_1=0, q_0=0$
1	001	$q_2=0, q_1=0, q_0=1$
2	010	$q_2=0, q_1=1, q_0=0$
3	011	$q_2=0, q_1=1, q_0=1$
4	100	$q_2=1, q_1=0, q_0=0$
5	101	$q_2=1, q_1=0, q_0=1$
6	110	$q_2=1, q_1=1, q_0=0$
7	111	$q_2=1, q_1=1, q_0=1$

Initial state: $|000\rangle = [1, 0, 0, 0, 0, 0, 0, 0]$

Circuit — 3 qubits, depth 4, with gates across each layer:

Layer	Gates
1	$H(q_0)$, $X(q_1)$, $RY(\pi/3)(q_2)$
2	$CNOT(q_0 \rightarrow q_1)$, $RZ(\pi/4)(q_2)$
3	$CNOT(q_1 \rightarrow q_2)$, $S(q_0)$
4	$RX(\pi/6)(q_0)$, $T(q_1)$, $H(q_2)$

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Gate Matrices Used

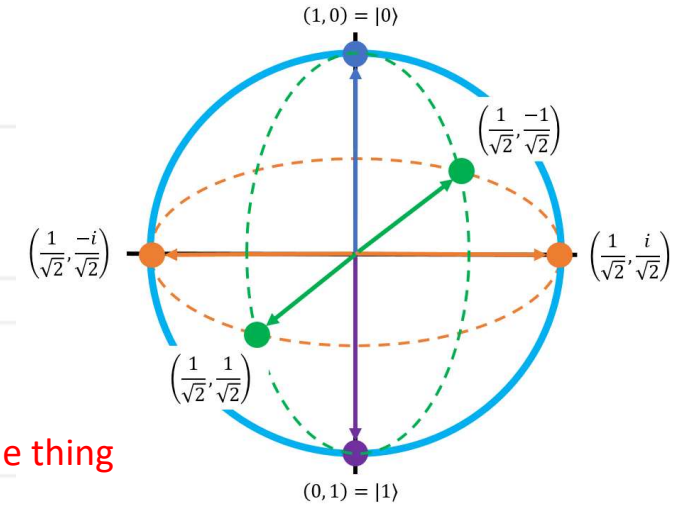
Hadamard (H):

```
H = (1/√2) [ 1   1 ]      H|0⟩ = |+⟩ = (|0⟩+|1⟩)/√2
           [ 1  -1 ]      H|1⟩ = |-⟩ = (|0⟩-|1⟩)/√2
```

Pauli-X:

```
X = [ 0  1 ]              X|0⟩ = |1⟩
    [ 1  0 ]              X|1⟩ = |0⟩
```

the Pauli-X gate and the X gate are the same thing



RY(θ):

```
RY(θ) = [ cos(θ/2)  -sin(θ/2) ]     RY(π/3)|0⟩ = cos(π/6)|0⟩ + sin(π/6)|1⟩
        [ sin(θ/2)   cos(θ/2) ]                = (√3/2)|0⟩ + (1/2)|1⟩
```

RZ(θ):

```
RZ(θ) = [ e^{-iθ/2}    0        ]    Rotates phase around Z-axis
        [    0       e^{iθ/2}    ]
```

$$R_x(\theta) = \begin{bmatrix} \cos(\theta/2) & -i\sin(\theta/2) \\ -i\sin(\theta/2) & \cos(\theta/2) \end{bmatrix}$$

CNOT (CX): If control qubit = |1⟩, flip the target qubit.

S gate: RZ(π/2) up to global phase → adds phase i to |1⟩ component.

T gate: RZ(π/4) up to global phase → adds phase e^{iπ/4} to |1⟩ component.

S gate: |0⟩ → |0⟩, |1⟩ → i|1⟩

T: |0⟩ → |0⟩, |1⟩ → $e^{i\pi/4}$|1⟩; e^{iπ/4} ≈ 0.7071 + 0.7071i

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Bloch Sphere Coordinates

For a single qubit described by a 2×2 **density matrix** ρ:

$$\rho = \begin{bmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{bmatrix}$$

$x = 2 \cdot \mathrm{Re}(\rho_{01})$ → measures superposition on X-axis

$y = 2 \cdot \mathrm{Im}(\rho_{10})$ → measures phase on Y-axis

$z = \rho_{00} - \rho_{11}$ → measures $|0\rangle$ vs $|1\rangle$ population

$r = \sqrt{x^2 + y^2 + z^2}$ → r=1 means pure state, r<1 means mixed (entangled)

For n-qubits, the state is represented as a tensor product of individual qubit states:

$$|\psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle \otimes \cdots \otimes |\psi_n\rangle.$$

$$|q\rangle = |q_1\rangle \otimes |q_2\rangle \otimes \cdots \otimes |q_n\rangle.$$

Tensor product of qubits:

$$|\psi\rangle = |0\rangle \otimes |1\rangle = |01\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}.$$

Tensor product of individual states:

$$|q\rangle = |q_1\rangle \otimes |q_2\rangle = \alpha\gamma|00\rangle + \alpha\delta|01\rangle + \beta\gamma|10\rangle + \beta\delta|11\rangle$$

$$|q_1\rangle = \alpha|0\rangle + \beta|1\rangle$$
$$|q_2\rangle = \gamma|0\rangle + \delta|1\rangle$$

Vector form:

$$|q\rangle = \begin{bmatrix} \alpha\gamma \\ \alpha\delta \\ \beta\gamma \\ \beta\delta \end{bmatrix}.$$

Density matrix ของระบบทั้งหมด คือ statevector คูณกับตัวเอง: $\rho = |\psi\rangle\langle\psi|$

สมมติ 2 qubits มี statevector 4 ตัว:

$$|\psi\rangle = [a, b, c, d]^T$$

$$\rho = |\psi\rangle\langle\psi| = \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} \times \begin{bmatrix} a^* & b^* & c^* & d^* \end{bmatrix}$$

$$= \begin{bmatrix} aa^* & ab^* & ac^* & ad^* \\ ba^* & bb^* & bc^* & bd^* \\ ca^* & cb^* & cc^* & cd^* \\ da^* & db^* & dc^* & dd^* \end{bmatrix} \quad \leftarrow \text{matrix } 4\times4$$

Density Matrix ของ Qubit เดียว

เริ่มจากสถานะทั่วไปของ 1 qubit

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

โดยที่ $|\alpha|^2 + |\beta|^2 = 1$

สร้าง Density Matrix: $\rho = |\psi\rangle\langle\psi|$

$$\rho = |\psi\rangle\langle\psi| = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \times \begin{bmatrix} \alpha^* & \beta^* \end{bmatrix}$$

$$= \begin{bmatrix} \alpha\alpha^* & \alpha\beta^* \\ \beta\alpha^* & \beta\beta^* \end{bmatrix} = \begin{bmatrix} |\alpha|^2 & \alpha\beta^* \\ \beta\alpha^* & |\beta|^2 \end{bmatrix}$$

$\cos(x)$ periodicity table with $2\pi n$ cycle:

x	$\cos(x)$	x	$\cos(x)$
0	1	π	-1
$\frac{\pi}{6}$	$\frac{\sqrt{3}}{2}$	$\frac{7\pi}{6}$	$-\frac{\sqrt{3}}{2}$
$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{5\pi}{4}$	$-\frac{\sqrt{2}}{2}$
$\frac{\pi}{3}$	$\frac{1}{2}$	$\frac{4\pi}{3}$	$-\frac{1}{2}$
$\frac{\pi}{2}$	0	$\frac{3\pi}{2}$	0
$\frac{2\pi}{3}$	$-\frac{1}{2}$	$\frac{5\pi}{3}$	$\frac{1}{2}$
$\frac{3\pi}{4}$	$-\frac{\sqrt{2}}{2}$	$\frac{7\pi}{4}$	$\frac{\sqrt{2}}{2}$
$\frac{5\pi}{6}$	$-\frac{\sqrt{3}}{2}$	$\frac{11\pi}{6}$	$\frac{\sqrt{3}}{2}$

$\sin(x)$ periodicity table with $2\pi n$ cycle:

x	$\sin(x)$	x	$\sin(x)$
0	0	π	0
$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{7\pi}{6}$	$-\frac{1}{2}$
$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{5\pi}{4}$	$-\frac{\sqrt{2}}{2}$
$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{4\pi}{3}$	$-\frac{\sqrt{3}}{2}$
$\frac{\pi}{2}$	1	$\frac{3\pi}{2}$	-1
$\frac{2\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{5\pi}{3}$	$-\frac{\sqrt{3}}{2}$
$\frac{3\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{7\pi}{4}$	$-\frac{\sqrt{2}}{2}$
$\frac{5\pi}{6}$	$\frac{1}{2}$	$\frac{11\pi}{6}$	$-\frac{1}{2}$

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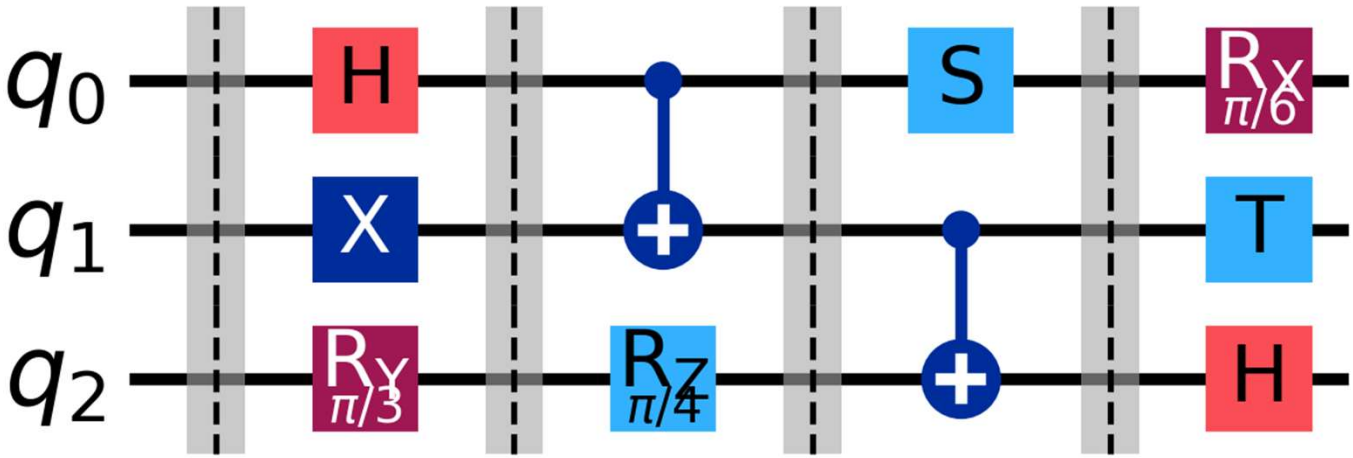
Full Quantum Circuit (depth = 4)



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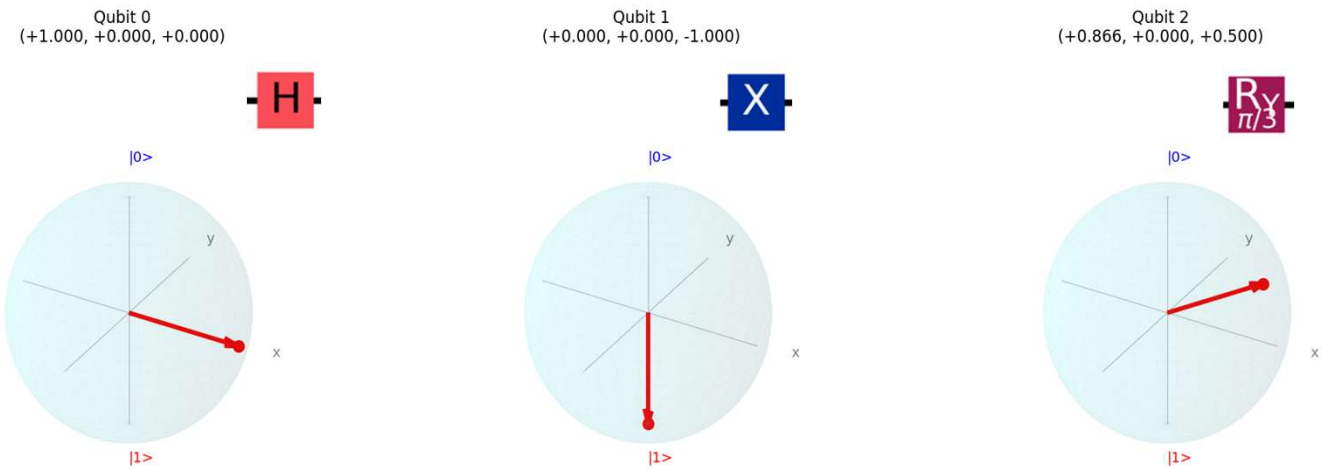
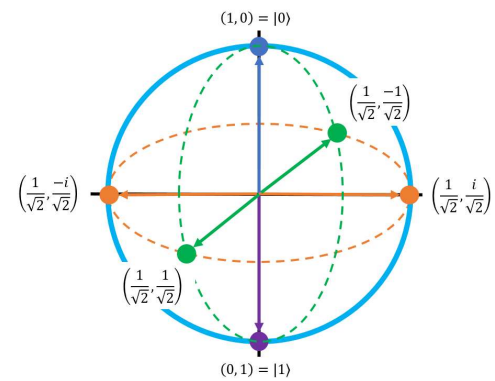
After Depth Layer 1





Input

0>
0>
0>

Layer 1: H(q_0), X(q_1), RY(π/3)(q_2)

All gates act on separate qubits (no entanglement), so the total state is a **tensor product**:

Individual qubit states:

q_0: $H|0\rangle = (1/\sqrt{2})|0\rangle + (1/\sqrt{2})|1\rangle$

q_1: $X|0\rangle = |1\rangle$

q_2: $RY(\pi/3)|0\rangle = \cos(\pi/6)|0\rangle + \sin(\pi/6)|1\rangle$
$= (\sqrt{3}/2)|0\rangle + (1/2)|1\rangle$

Total state = $|q_2\rangle \otimes |q_1\rangle \otimes |q_0\rangle$:

$= [(\sqrt{3}/2)|0\rangle + (1/2)|1\rangle] \otimes |1\rangle \otimes [(1/\sqrt{2})|0\rangle + (1/\sqrt{2})|1\rangle]$

Expanding (multiply out all combinations):

$(\sqrt{3}/2)\cdot(1)\cdot(1/\sqrt{2})\ |010\rangle = \sqrt{6}/4 \approx 0.6124\ |010\rangle$
$(\sqrt{3}/2)\cdot(1)\cdot(1/\sqrt{2})\ |011\rangle = \sqrt{6}/4 \approx 0.6124\ |011\rangle$
$(1/2)\cdot(1)\cdot(1/\sqrt{2})\ |110\rangle = \sqrt{2}/4 \approx 0.3536\ |110\rangle$
$(1/2)\cdot(1)\cdot(1/\sqrt{2})\ |111\rangle = \sqrt{2}/4 \approx 0.3536\ |111\rangle$

Bloch coordinates (no entanglement, so each qubit is pure → r = 1):

$q_0 = |+\rangle$: $\rho = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$ → x=1, y=0, z=0 r=1

$q_1 = |1\rangle$: $\rho = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ → x=0, y=0, z=-1 r=1

$q_2 = (\sqrt{3}/2)|0\rangle + (1/2)|1\rangle$:
$\rho = \begin{bmatrix} 3/4 & \sqrt{3}/4 \\ \sqrt{3}/4 & 1/4 \end{bmatrix}$ → x=√3/2≈0.866, y=0, z=1/2 r=1

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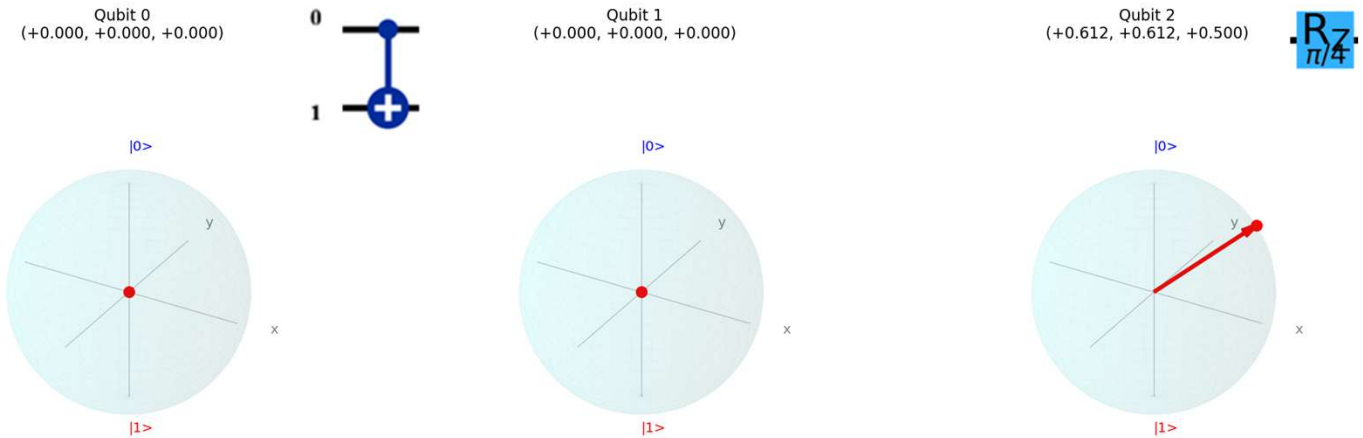
After Depth Layer 2




```
Statevector  (layer 2):

  |001>  ->  +0.5658-0.2343j   (P=0.3750)
  |010>  ->  +0.5658-0.2343j   (P=0.3750)
  |101>  ->  +0.3266+0.1353j   (P=0.1250)
  |110>  ->  +0.3266+0.1353j   (P=0.1250)

Bloch coords (x, y, z):
  q0: (+0.0000, +0.0000, +0.0000)  r=0.0000
  q1: (+0.0000, +0.0000, +0.0000)  r=0.0000
  q2: (+0.6124, +0.6124, +0.5000)  r=1.0000
```

From Layer1

```
|010⟩  =  √6/4
|011⟩  =  √6/4
|110⟩  =  √2/4
|111⟩  =  √2/4
```

CNOT with control=q_0, target=q_1

Rule: if q_0=1, flip q_1. Apply to each basis state:

```
|010⟩ : q₀=0 → no flip   → |010⟩
|011⟩ : q₀=1 → flip q₁   → |001⟩   (q₁: 1→0)
|110⟩ : q₀=0 → no flip   → |110⟩
|111⟩ : q₀=1 → flip q₁   → |101⟩   (q₁: 1→0)
```

After CX:

```
√6/4 |010⟩ + √6/4 |001⟩ + √2/4 |110⟩ + √2/4 |101⟩
```

RZ adds a phase depending on q_2's value:

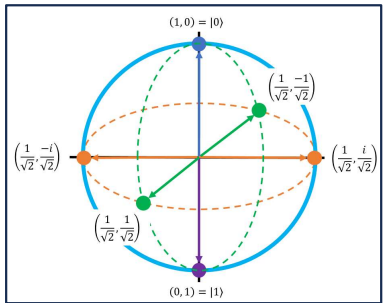
```
q₂ = |0⟩ → multiply by e^{-iπ/8}
q₂ = |1⟩ → multiply by e^{+iπ/8}
```

Where:

```
e^{-iπ/8} = cos(π/8) - i·sin(π/8) ≈ 0.9239 - 0.3827i
e^{+iπ/8} = cos(π/8) + i·sin(π/8) ≈ 0.9239 + 0.3827i
```

Apply to each term:

```
|001⟩ (q₂=0): √6/4 × e^{-iπ/8} = 0.6124 × (0.9239-0.3827i) ≈ 0.5658 - 0.2343i
|010⟩ (q₂=0): √6/4 × e^{-iπ/8} ≈ 0.5658 - 0.2343i
|101⟩ (q₂=1): √2/4 × e^{+iπ/8} = 0.3536 × (0.9239+0.3827i) ≈ 0.3266 + 0.1353i
|110⟩ (q₂=1): √2/4 × e^{+iπ/8} ≈ 0.3266 + 0.1353i
```




```
RZ(θ) = [ e^{-iθ/2}     0        ]
        [     0       e^{iθ/2}   ]
```

Bloch coordinates:

```
q₀: (0, 0, 0)  r=0  → maximally mixed (center of sphere)
q₁: (0, 0, 0)  r=0  → maximally mixed (center of sphere)
q₂: (+0.6124, +0.6124, +0.5000)  r=1  → still pure
```

LAYER 2: CNOT($q_0 \to q_1$), Rz($\pi/4$)(q_2)

This layer introduces the first entangling operation. The CNOT has q_0 as control and q_1 as target.

CNOT($q_0 \to q_1$) Action

The CNOT flips q_1 whenever $q_0 = |1\rangle$:

$$|010\rangle \xrightarrow{\text{CNOT}} |010\rangle \quad (q_0 = 0,\ \text{no flip})$$

$$|011\rangle \xrightarrow{\text{CNOT}} |011\rangle \quad (q_0 = 0,\ \text{no flip})$$

$$|110\rangle \xrightarrow{\text{CNOT}} |100\rangle \quad (q_0 = 1,\ \text{flip } q_1 : 1 \to 0)$$

$$|111\rangle \xrightarrow{\text{CNOT}} |101\rangle \quad (q_0 = 1,\ \text{flip } q_1 : 1 \to 0)$$

After CNOT (before Rz):

$$|\psi'\rangle = \frac{\sqrt{3}}{2\sqrt{2}}|010\rangle + \frac{1}{2\sqrt{2}}|011\rangle + \frac{\sqrt{3}}{2\sqrt{2}}|100\rangle + \frac{1}{2\sqrt{2}}|101\rangle$$

Rz($\pi/4$) on q_2

$$R_z(\pi/4) = \begin{pmatrix} e^{-i\pi/8} & 0 \\ 0 & e^{i\pi/8} \end{pmatrix}$$

This applies a phase e^(−iπ/8) to |0⟩ components of q_2 and e^(+iπ/8) to |1⟩ components:

Term	q_2	Phase applied	Result
|010⟩	0	e^(−iπ/8)	e^(−iπ/8) · √3/(2√2)
|011⟩	1	e^(+iπ/8)	e^(+iπ/8) · 1/(2√2)
|100⟩	0	e^(−iπ/8)	e^(−iπ/8) · √3/(2√2)
|101⟩	1	e^(+iπ/8)	e^(+iπ/8) · 1/(2√2)

Full 3-Qubit State After Layer 2

$$|\psi_2\rangle = \frac{\sqrt{3}}{2\sqrt{2}}e^{-i\pi/8}|010\rangle + \frac{1}{2\sqrt{2}}e^{i\pi/8}|011\rangle + \frac{\sqrt{3}}{2\sqrt{2}}e^{-i\pi/8}|100\rangle + \frac{1}{2\sqrt{2}}e^{i\pi/8}|101\rangle$$

Using e^(−iπ/8) = cos(π/8) − i·sin(π/8) ≈ 0.9239 − 0.3827i:

Basis State	Amplitude	|c|	Phase
|010⟩	0.5658 − 0.2343i	0.6124	−π/8
|011⟩	0.3266 + 0.1353i	0.3536	+π/8
|100⟩	0.5658 − 0.2343i	0.6124	−π/8
|101⟩	0.3266 + 0.1353i	0.3536	+π/8

L2

Single-Qubit Analysis After Layer 2

q_0 — Now ENTANGLED:

$$\rho_{q_0} = \mathrm{Tr}_{q_1,q_2}(|\psi_2\rangle\langle\psi_2|) = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/2 \end{pmatrix} = \frac{I}{2}$$

Bloch vector: (0, 0, 0) — at the **center** of the Bloch sphere! **r = 0** → maximally mixed single-qubit state.

q_1 — Also ENTANGLED:

$$\rho_{q_1} = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/2 \end{pmatrix} = \frac{I}{2}$$

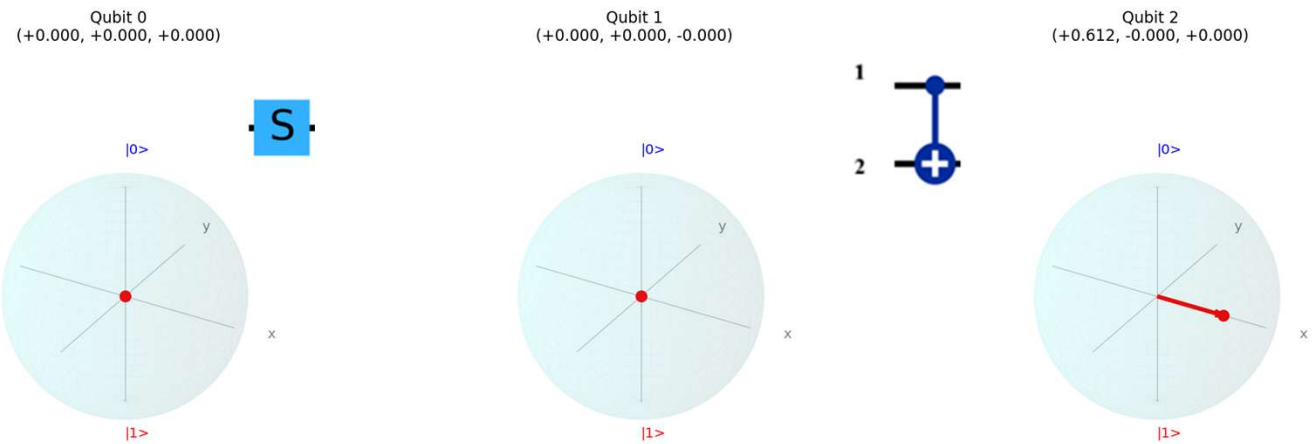
Bloch vector: (0, 0, 0), **r = 0, S(q_1) = 1.0 bit.** Maximally mixed — mirror image of q_0's entanglement.

q_2 — Still PURE (unentangled):

$$\rho_{q_2} = \begin{pmatrix} 3/4 & \frac{\sqrt{3}}{4}e^{-i\pi/4} \\ \frac{\sqrt{3}}{4}e^{i\pi/4} & 1/4 \end{pmatrix} = \begin{pmatrix} 0.7500 & 0.3062 - 0.3062i \\ 0.3062 + 0.3062i & 0.2500 \end{pmatrix}$$

Bloch vector: ($\sqrt{3}/2$, $\sqrt{3}/2$, 1/2) ≈ (0.6124, 0.6124, 0.5000), **r = 1.0.**

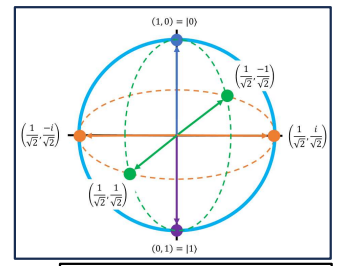
After Depth Layer 3




```
Statevector  (layer 3):

  |001>  ->  +0.2343+0.5658j   (P=0.3750)
  |010>  ->  +0.3266+0.1353j   (P=0.1250)
  |101>  ->  -0.1353+0.3266j   (P=0.1250)
  |110>  ->  +0.5658-0.2343j   (P=0.3750)

Bloch coords (x, y, z):
  q0: (+0.0000, +0.0000, +0.0000)  r=0.0000
  q1: (+0.0000, +0.0000, -0.0000)  r=0.0000
  q2: (+0.6124, -0.0000, +0.0000)  r=0.6124
```



S gate: $|0\rangle \to |0\rangle$, $|1\rangle \to i|1\rangle$

```
|001⟩ (q₀=1):  × i  →  i(0.5658-0.2343i) = 0.2343 + 0.5658i
|010⟩ (q₀=0):  × 1  →  0.3266 + 0.1353i
|101⟩ (q₀=1):  × i  →  i(0.3266+0.1353i) = -0.1353 + 0.3266i
|110⟩ (q₀=0):  × 1  →  0.5658 - 0.2343i
```

From Layer2

$|010\rangle = 0.5658 - 0.2343i$

$|011\rangle = 0.3266 + 0.1353i$

$|100\rangle = 0.5658 - 0.2343i$

$|101\rangle = 0.3266 + 0.1353i$

CNOT with control=q_1, target=q_2

```
|001⟩ : q₁=0 → no flip  → |001⟩
|010⟩ : q₁=1 → flip q₂  → |110⟩   (q₂: 0→1)
|101⟩ : q₁=0 → no flip  → |101⟩
|110⟩ : q₁=1 → flip q₂  → |010⟩   (q₂: 1→0)
```

```
(0.5658-0.2343i)|001⟩     (q₁=0, no flip)
(0.5658-0.2343i)|110⟩     (q₁=1, flip q₂: 0→1)
(0.3266+0.1353i)|101⟩     (q₁=0, no flip)
(0.3266+0.1353i)|010⟩     (q₁=1, flip q₂: 1→0)
```

```
|001⟩:  0.5658 - 0.2343i
|010⟩:  0.3266 + 0.1353i
|101⟩:  0.3266 + 0.1353i
|110⟩:  0.5658 - 0.2343i
```

Bloch coordinates:

```
q₀: (0, 0, 0)          r=0       → still maximally mixed
q₁: (0, 0, 0)          r≈0       → still maximally mixed
q₂: (+0.6124, 0, 0)    r=0.6124  → partially mixed
```

LAYER 3: $S(q_0)$, $CNOT(q_1 \to q_2)$

Step 1: S Gate on q_0

S adds a phase of i to the $|1\rangle$ component of q_0:

Term	q_0	Phase	Result
$\|010\rangle$	0	1	unchanged
$\|011\rangle$	0	1	unchanged
$\|100\rangle$	1	i	multiplied by i
$\|101\rangle$	1	i	multiplied by i

Step 2: $CNOT(q_1 \to q_2)$

Now q_1 is control, q_2 is target. Flip q_2 when $q_1 = |1\rangle$:

Before	q_1	Action	After
$\|010\rangle$	1	flip q_2	$\|011\rangle$
$\|011\rangle$	1	flip q_2	$\|010\rangle$
$\|100\rangle$	0	no flip	$\|100\rangle$
$\|101\rangle$	0	no flip	$\|101\rangle$

Full 3-Qubit State After Layer 3

$$|\psi_3\rangle = \frac{1}{2\sqrt{2}}e^{i\pi/8}|010\rangle + \frac{\sqrt{3}}{2\sqrt{2}}e^{-i\pi/8}|011\rangle + \frac{i\sqrt{3}}{2\sqrt{2}}e^{-i\pi/8}|100\rangle + \frac{i}{2\sqrt{2}}e^{i\pi/8}|101\rangle$$

Basis State	Amplitude	|c|	Phase
|010⟩	0.3266 + 0.1353i	0.3536	π/8
|011⟩	0.5658 − 0.2343i	0.6124	−π/8
|100⟩	0.2343 + 0.5658i	0.6124	3π/8
|101⟩	−0.1353 + 0.3266i	0.3536	5π/8

Single-Qubit Analysis After Layer 3

q_0 — Still maximally entangled:

$$\rho_{q_0} = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/2 \end{pmatrix}$$

Bloch: (0, 0, 0), r = 0, S = 1.0 bit. The S gate only added a relative phase within the entangled state — it did not break or change the entanglement structure.

q_1 — Still maximally entangled:

$$\rho_{q_1} = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/2 \end{pmatrix}$$

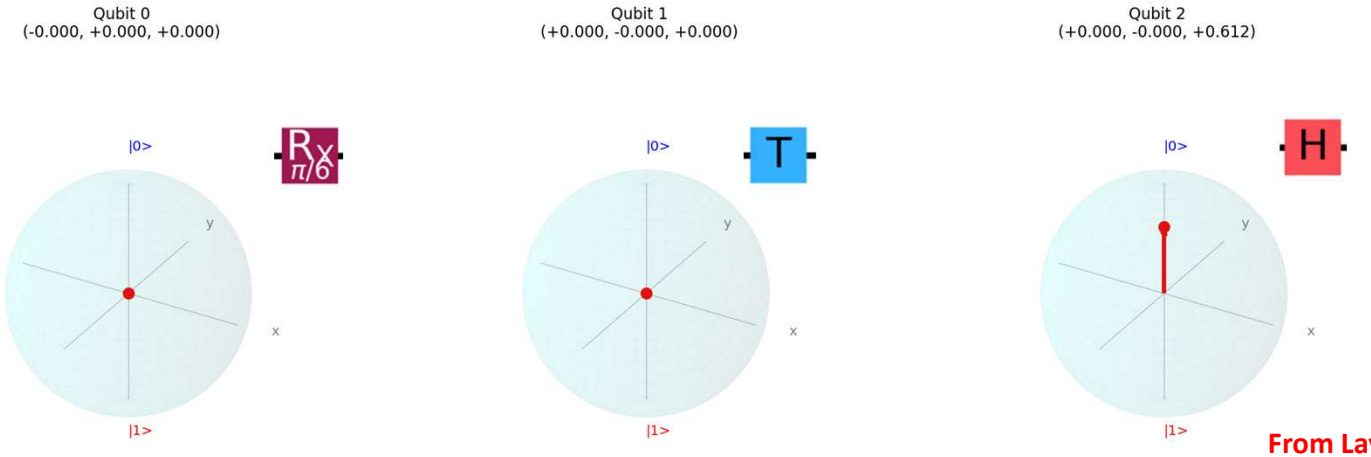
Bloch: (0, 0, 0), r = 0, S = 1.0 bit.

q_2 — Now PARTIALLY ENTANGLED:

$$\rho_{q_2} = \begin{pmatrix} 1/2 & \sqrt{3}/2\sqrt{2} \\ \sqrt{3}/2\sqrt{2} & 1/2 \end{pmatrix} = \begin{pmatrix} 0.5000 & 0.3062 \\ 0.3062 & 0.5000 \end{pmatrix}$$

Bloch vector: $(\sqrt{3}/2\sqrt{2}, 0, 0) \approx (0.6124, 0, 0)$, **r ≈ 0.6124, $S(q_2)$ ≈ 0.7094 bits.**

After Depth Layer 4




```
Statevector  (layer 4):

  |000>  ->  +0.1633-0.0181j   (P=0.0270)
  |001>  ->  +0.0676+0.6095j   (P=0.3761)
  |010>  ->  +0.4788+0.3832j   (P=0.3761)
  |011>  ->  +0.1027-0.1283j   (P=0.0270)
  |100>  ->  +0.0438-0.0676j   (P=0.0065)
  |101>  ->  +0.2525+0.1633j   (P=0.0904)
  |110>  ->  -0.2940+0.0630j   (P=0.0904)
  |111>  ->  +0.0169+0.0788j   (P=0.0065)

Bloch coords (x, y, z):
  q0: (-0.0000, +0.0000, +0.0000)  r=0.0000
  q1: (+0.0000, -0.0000, +0.0000)  r=0.0000
  q2: (+0.0000, -0.0000, +0.6124)  r=0.6124
```

From Layer3

$$|010\rangle = 0.3266 + 0.1353i$$

$$|011\rangle = 0.5658 - 0.2343i$$

$$|100\rangle = 0.2343 + 0.5658i$$

$$|101\rangle = -0.1353 + 0.3266i$$

RX(π/6) on q_0

```
RX(θ) = [ cos(θ/2)    -i·sin(θ/2) ]
        [ -i·sin(θ/2)  cos(θ/2)    ]

RX(π/6): cos(π/12) ≈ 0.9659,  sin(π/12) ≈ 0.2588
```

For each basis state, RX acts on the q_0 part:

```
|..0⟩ → cos(π/12)|..0⟩  - i·sin(π/12)|..1⟩
|..1⟩ → -i·sin(π/12)|..0⟩ + cos(π/12)|..1⟩
```

T gate on q_1

T: $|0\rangle \to |0\rangle$, $|1\rangle \to e^{i\pi/4}|1\rangle$ where $e^{i\pi/4} \approx 0.7071 + 0.7071i$

Adds phase to all states where q_1=1.

H gate on q_2

```
|0⟩ → (|0⟩ + |1⟩)/√2
|1⟩ → (|0⟩ - |1⟩)/√2
```

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The Three Gate Matrices

Rx(π/6) — Rotation around X-axis by 30°

With θ = π/6, the half-angle is θ/2 = π/12 = 15°:

$$\cos(\pi/12) = \frac{\sqrt{6}+\sqrt{2}}{4} \approx 0.965926$$

$$\sin(\pi/12) = \frac{\sqrt{6}-\sqrt{2}}{4} \approx 0.258819$$

$$R_x(\pi/6) = \begin{pmatrix} \cos(\pi/12) & -i\sin(\pi/12) \\ -i\sin(\pi/12) & \cos(\pi/12) \end{pmatrix} = \begin{pmatrix} 0.9659 & -0.2588i \\ -0.2588i & 0.9659 \end{pmatrix}$$

Action on basis states:

$$R_x(\pi/6)|0\rangle = \cos(\pi/12)|0\rangle - i\sin(\pi/12)|1\rangle = 0.9659|0\rangle - 0.2588i|1\rangle$$

$$R_x(\pi/6)|1\rangle = -i\sin(\pi/12)|0\rangle + \cos(\pi/12)|1\rangle = -0.2588i|0\rangle + 0.9659|1\rangle$$

T — Phase gate (π/4 rotation around Z-axis)

$$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1+i}{\sqrt{2}} \end{pmatrix}$$

Action on basis states:

$$T|0\rangle = |0\rangle \quad \text{(unchanged)}$$

$$T|1\rangle = e^{i\pi/4}|1\rangle = \frac{1+i}{\sqrt{2}}|1\rangle \approx (0.7071 + 0.7071i)|1\rangle$$

H — Hadamard

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

Action on basis states:

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

L4

The Combined Operator

Layer 4 is a **local (separable) unitary**:

$$U_4 = R_x(\pi/6) \otimes T \otimes H$$

This is an 8×8 matrix, but because it's a tensor product, we can apply each gate independently to its respective qubit. Each input basis state $|q_0\, q_1\, q_2\rangle$ transforms as:

$$U_4|q_0 q_1 q_2\rangle = \big(R_x(\pi/6)|q_0\rangle\big) \otimes \big(T|q_1\rangle\big) \otimes \big(H|q_2\rangle\big)$$

(0.3266 + 0.1353i)|010⟩

$\|010\rangle$	=	0.3266 + 0.1353i
$\|011\rangle$	=	0.5658 − 0.2343i
$\|100\rangle$	=	0.2343 + 0.5658i
$\|101\rangle$	=	−0.1353 + 0.3266i

Decompose $|010\rangle = |0\rangle_{(q_0)} \otimes |1\rangle_{(q_1)} \otimes |0\rangle_{(q_2)}$

Rx(π/6)|0⟩:

$$= 0.9659|0\rangle - 0.2588i|1\rangle$$

T|1⟩:

$$= e^{i\pi/4}|1\rangle = (0.7071 + 0.7071i)|1\rangle$$

q_1 stays in $|1\rangle$ but acquires a 45° phase rotation.

H|0⟩:

$$= \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

Tensor product (multiplied by the original coefficient 0.3266 + 0.1353i):

Output State	Amplitude
$\|010\rangle$	+0.0924 + 0.2231i
$\|011\rangle$	+0.0924 + 0.2231i
$\|110\rangle$	+0.0598 − 0.0248i
$\|111\rangle$	+0.0598 − 0.0248i

L4

$\|010\rangle$	=	0.3266 + 0.1353i
$\|011\rangle$	=	0.5658 − 0.2343i
$\|100\rangle$	=	0.2343 + 0.5658i
$\|101\rangle$	=	−0.1353 + 0.3266i

(0.5658 − 0.2343i)|011⟩

Decompose $|011\rangle = |0\rangle_{(q_0)} \otimes |1\rangle_{(q_1)} \otimes |1\rangle_{(q_2)}$

Rx(π/6)|0⟩:

$$= 0.9659|0\rangle - 0.2588i|1\rangle$$

T|1⟩:

$$= (0.7071 + 0.7071i)|1\rangle$$

H|1⟩:

$$= \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

Tensor product (multiplied by 0.5658 − 0.2343i):

Output State	Amplitude
$\|010\rangle$	+0.3864 + 0.1601i
$\|011\rangle$	**−0.3864 − 0.1601i** ← sign flip from H
$\|110\rangle$	+0.0429 − 0.1035i
$\|111\rangle$	**−0.0429 + 0.1035i** ← sign flip from H

$\|010\rangle$	=	$0.3266 + 0.1353i$
$\|011\rangle$	=	$0.5658 - 0.2343i$
$\|100\rangle$	=	$0.2343 + 0.5658i$
$\|101\rangle$	=	$-0.1353 + 0.3266i$

L4

(0.2343 + 0.5658i)|100⟩

Decompose $|100\rangle = |1\rangle_{(q_0)} \otimes |0\rangle_{(q_1)} \otimes |0\rangle_{(q_2)}$

Rx(π/6)|1⟩:

$$= -0.2588i|0\rangle + 0.9659|1\rangle$$

T|0⟩:

$$= |0\rangle \quad \text{(unchanged — T has no effect on } |0\rangle\text{)}$$

H|0⟩:

$$= \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

Tensor product (multiplied by 0.2343 + 0.5658i):

Output State	Amplitude
$\|000\rangle$	+0.1035 − 0.0429i
$\|001\rangle$	+0.1035 − 0.0429i
$\|100\rangle$	+0.1601 + 0.3864i
$\|101\rangle$	+0.1601 + 0.3864i

L4

$\|010\rangle$	=	$0.3266 + 0.1353i$
$\|011\rangle$	=	$0.5658 - 0.2343i$
$\|100\rangle$	=	$0.2343 + 0.5658i$
$\|101\rangle$	=	$-0.1353 + 0.3266i$

$(-0.1353 + 0.3266i)|101\rangle$

Decompose $|101\rangle = |1\rangle_{(q_0)} \otimes |0\rangle_{(q_1)} \otimes |1\rangle_{(q_2)}$

$Rx(\pi/6)|1\rangle$:

$$= -0.2588i|0\rangle + 0.9659|1\rangle$$

$T|0\rangle$:

$$= |0\rangle$$

$-H|1\rangle$:

$$= \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \quad \leftarrow \text{minus sign again!}$$

Tensor product (multiplied by $-0.1353 + 0.3266i$):

Output State	Amplitude
$\|000\rangle$	$+0.0598 + 0.0248i$
$\|001\rangle$	**$-0.0598 - 0.0248i$**
$\|100\rangle$	$-0.0924 + 0.2231i$
$\|101\rangle$	**$+0.0924 - 0.2231i$**

Quantum Interference — Collecting All Contributions

|000⟩: Constructive interference

$$\underbrace{(+0.1035 - 0.0429i)}_{\text{from } |100\rangle} + \underbrace{(+0.0598 + 0.0248i)}_{\text{from } |101\rangle} = +\mathbf{0.1633} - \mathbf{0.0181i}$$

|001⟩: Destructive interference

$$\underbrace{(+0.1035 - 0.0429i)}_{\text{from } |100\rangle} + \underbrace{(-0.0598 - 0.0248i)}_{\text{from } |101\rangle} = +\mathbf{0.0438} - \mathbf{0.0676i}$$

|010⟩: STRONG constructive interference ★

$$\underbrace{(+0.0924 + 0.2231i)}_{\text{from } |010\rangle} + \underbrace{(+0.3864 + 0.1601i)}_{\text{from } |011\rangle} = +\mathbf{0.4788} + \mathbf{0.3832i}$$

|011⟩: Partial destructive interference

$$\underbrace{(+0.0924 + 0.2231i)}_{\text{from } |010\rangle} + \underbrace{(-0.3864 - 0.1601i)}_{\text{from } |011\rangle} = -\mathbf{0.2940} + \mathbf{0.0630i}$$

|100⟩: STRONG constructive interference ★

$$\underbrace{(+0.1601 + 0.3864i)}_{\text{from } |100\rangle} + \underbrace{(-0.0924 + 0.2231i)}_{\text{from } |101\rangle} = +\mathbf{0.0676} + \mathbf{0.6095i}$$

|101⟩: Partial cancellation

$$\underbrace{(+0.1601 + 0.3864i)}_{\text{from } |100\rangle} + \underbrace{(+0.0924 - 0.2231i)}_{\text{from } |101\rangle} = +\mathbf{0.2525} + \mathbf{0.1633i}$$

|110⟩: Small constructive

$$\underbrace{(+0.0598 - 0.0248i)}_{\text{from } |010\rangle} + \underbrace{(+0.0429 - 0.1035i)}_{\text{from } |011\rangle} = +\mathbf{0.1027} - \mathbf{0.1283i}$$

|111⟩: Small destructive

$$\underbrace{(+0.0598 - 0.0248i)}_{\text{from } |010\rangle} + \underbrace{(-0.0429 + 0.1035i)}_{\text{from } |011\rangle} = +\mathbf{0.0169} + \mathbf{0.0788i}$$

State	Amplitude
$\|000\rangle$	0.1633 − 0.0181i
$\|001\rangle$	0.0438 − 0.0677i
$\|010\rangle$	**0.4788 + 0.3832i**
$\|011\rangle$	−0.2940 + 0.0630i
$\|100\rangle$	**0.0677 + 0.6095i**
$\|101\rangle$	0.2525 + 0.1633i
$\|110\rangle$	0.1027 − 0.1283i
$\|111\rangle$	0.0169 + 0.0788i

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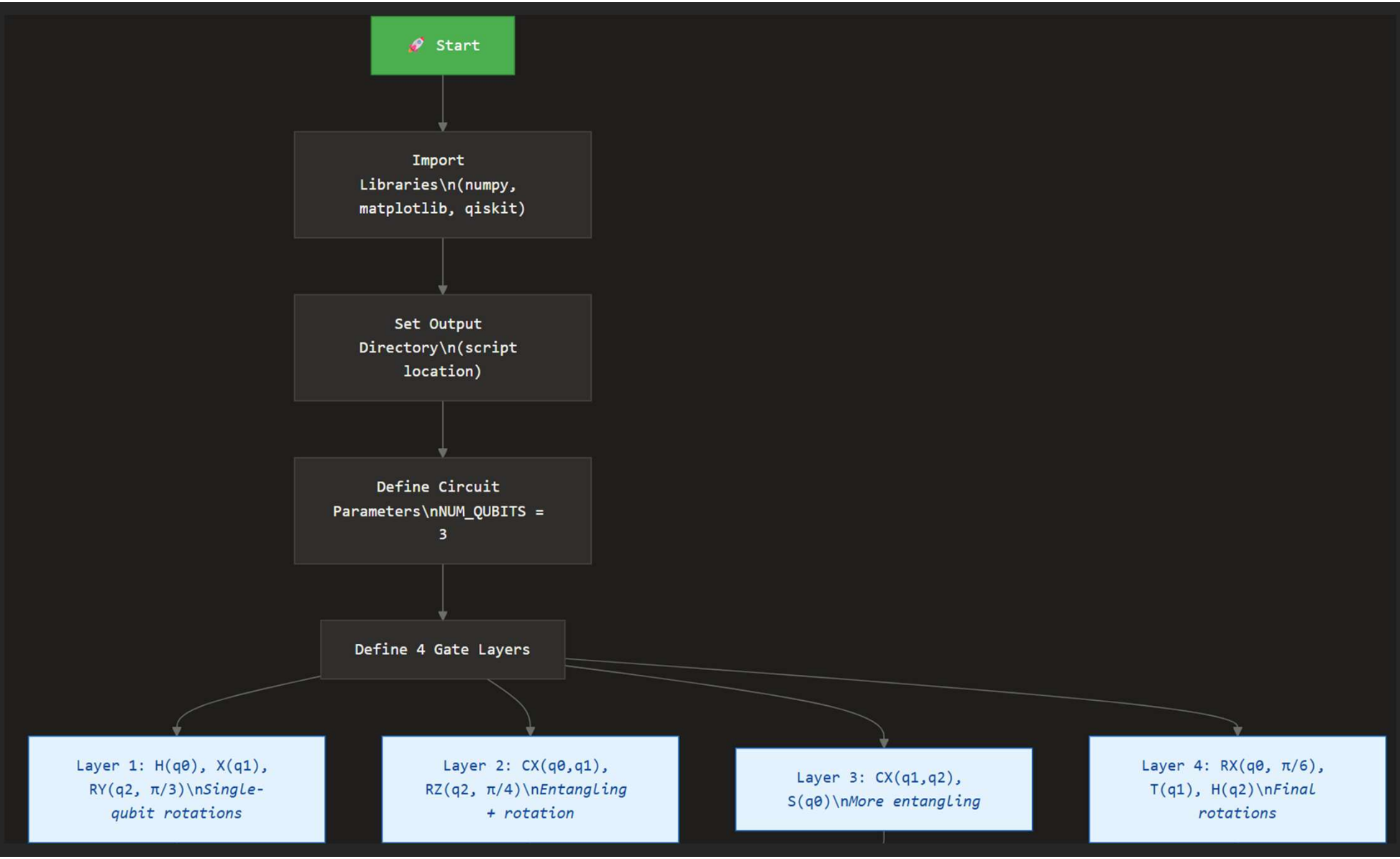
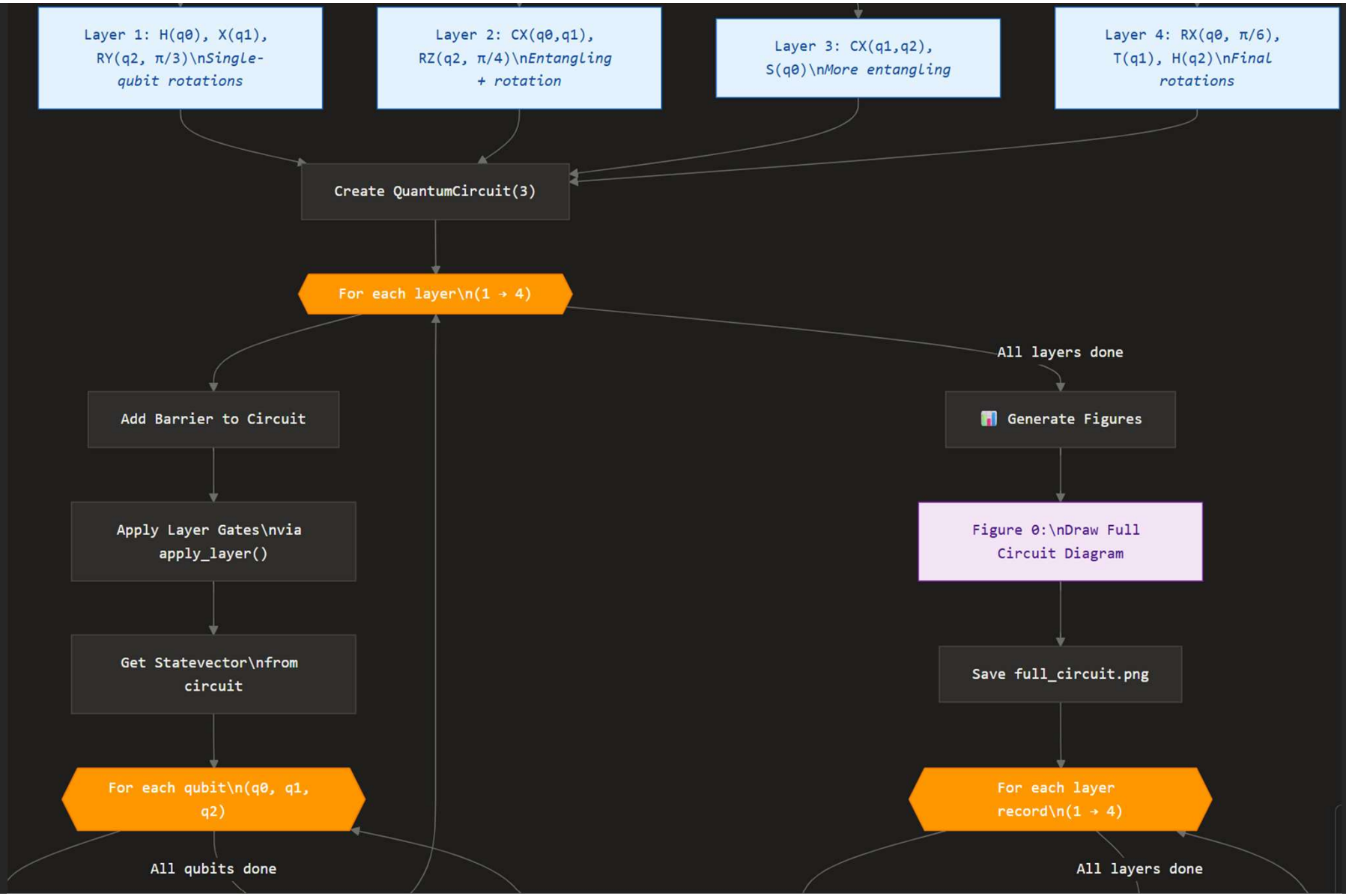
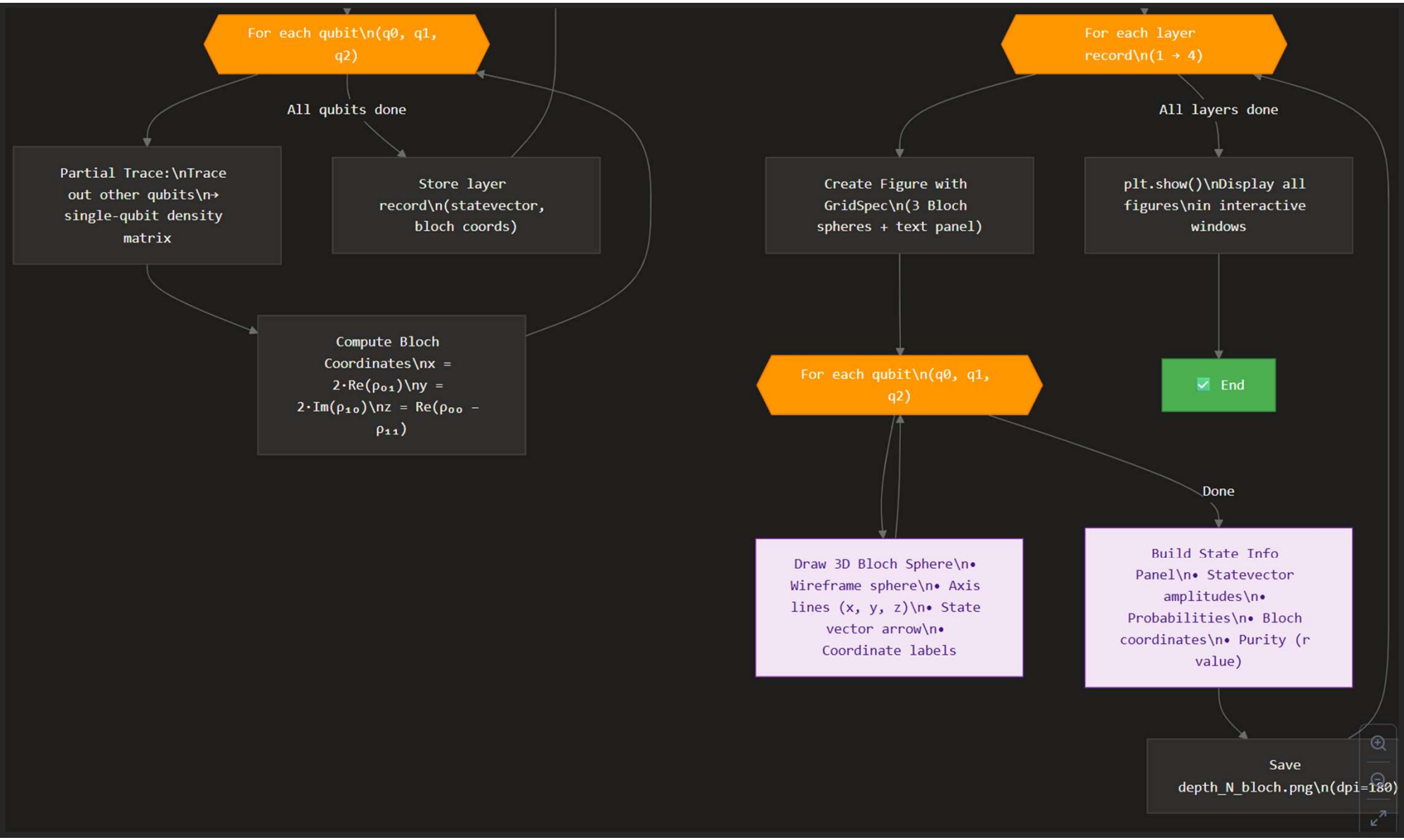

🚀 Start
Import Libraries\n(numpy, matplotlib, qiskit)
Set Output Directory\n(script location)
Define Circuit Parameters\nNUM_QUBITS = 3
Define 4 Gate Layers
Layer 1: H(q0), X(q1), RY(q2, π/3)\nSingle-qubit rotations
Layer 2: CX(q0,q1), RZ(q2, π/4)\nEntangling + rotation
Layer 3: CX(q1,q2), S(q0)\nMore entangling
Layer 4: RX(q0, π/6), T(q1), H(q2)\nFinal rotations


Layer 1: H(q0), X(q1), RY(q2, π/3)\nSingle-qubit rotations
Layer 2: CX(q0,q1), RZ(q2, π/4)\nEntangling + rotation
Layer 3: CX(q1,q2), S(q0)\nMore entangling
Layer 4: RX(q0, π/6), T(q1), H(q2)\nFinal rotations
Create QuantumCircuit(3)
For each layer\n(1 → 4)
All layers done
Add Barrier to Circuit
Apply Layer Gates\nvia apply_layer()
Get Statevector\nfrom circuit
For each qubit\n(q0, q1, q2)
All qubits done
📊 Generate Figures
Figure 0:\nDraw Full Circuit Diagram
Save full_circuit.png
For each layer record\n(1 → 4)
All layers done


For each qubit\n(q0, q1, q2)
All qubits done
Partial Trace:\nTrace out other qubits\n→ single-qubit density matrix
Store layer record\n(statevector, bloch coords)
Compute Bloch Coordinates\nx = 2·Re(ρ₀₁)\ny = 2·Im(ρ₁₀)\nz = Re(ρ₀₀ − ρ₁₁)
For each layer record\n(1 → 4)
All layers done
Create Figure with GridSpec\n(3 Bloch spheres + text panel)
plt.show()\nDisplay all figures\nin interactive windows
For each qubit\n(q0, q1, q2)
End
Done
Draw 3D Bloch Sphere\n• Wireframe sphere\n• Axis lines (x, y, z)\n• State vector arrow\n• Coordinate labels
Build State Info Panel\n• Statevector amplitudes\n• Probabilities\n• Bloch coordinates\n• Purity (r value)
Save depth_N_bloch.png\n(dpi=180)

```python
# -- Output directory (same folder as this script) ----------------------
OUT_DIR = os.path.dirname(os.path.abspath(__file__))

NUM_QUBITS = 3

layers = [
    # Layer 1 - single-qubit rotations
    [("h",  [0]),
     ("x",  [1]),
     ("ry", [2], np.pi / 3)],
    # Layer 2 - entangling + rotation
    [("cx", [0, 1]),
     ("rz", [2], np.pi / 4)],
    # Layer 3 - more entangling
    [("cx", [1, 2]),
     ("s",  [0])],
    # Layer 4 - final rotations
    [("rx", [0], np.pi / 6),
     ("t",  [1]),
     ("h",  [2])],
]
```

```python
# -- Simulate layer-by-layer ------------------------------------------->Start!
qc = QuantumCircuit(NUM_QUBITS)
records = []

for layer_idx, layer in enumerate(layers):
    print("------------ Front ------------------")
    print(layer_idx)
    print(layer)
    print("------------ Tails ------------------")
    qc.barrier()
    apply_layer(qc, layer)
    sv = Statevector.from_instruction(
        qc.remove_final_measurements(inplace=False) if qc.num_clbits else qc
    )
    layer_data = {"layer": layer_idx + 1, "statevector": sv, "qubits": []}
    for q in range(NUM_QUBITS):
        trace_out = [i for i in range(NUM_QUBITS) if i != q]
        dm = partial_trace(sv, trace_out)
        bx, by, bz = bloch_coords(dm.data)
        layer_data["qubits"].append({"qubit": q, "bloch": (bx, by, bz)})
    records.append(layer_data)
```

```python
def apply_layer(qc, layer):
    for instruction in layer:
        gate, qubits = instruction[0], instruction[1]
        param = instruction[2] if len(instruction) > 2 else None
        gate_map = {
            "h": qc.h, "x": qc.x, "y": qc.y, "z": qc.z,
            "s": qc.s, "t": qc.t,
        }
        param_gate_map = {"rx": qc.rx, "ry": qc.ry, "rz": qc.rz}
        two_q_map = {"cx": qc.cx, "cz": qc.cz}

        if gate in gate_map:
            gate_map[gate](*qubits)
        elif gate in param_gate_map:
            param_gate_map[gate](param, *qubits)
        elif gate in two_q_map:
            two_q_map[gate](*qubits)
```

1. `gate_map` — เกตที่ไม่มี parameter

python

```python
gate_map = {
    "h": qc.h,    # ← ไม่ได้เรียก qc.h() แค่อ้างถึง method
    "x": qc.x,
    "y": qc.y,
    "z": qc.z,
    "s": qc.s,
    "t": qc.t,
}
```

2. `param_gate_map` — เกตที่ต้องมีมุม (angle)

```python
param_gate_map = {"rx": qc.rx, "ry": qc.ry, "rz": qc.rz}
```

3. `two_q_map` — เกต 2 qubit

```python
two_q_map = {"cx": qc.cx, "cz": qc.cz}
```

```
print(qc.h) # <bound method QuantumCircuit.h of ...>
print(gate_map["h"]) # <bound method QuantumCircuit.h of ...> ← เหมือนกัน!
```

```
gate_map["h"]        # คือสิ่งเดียวกันกับ qc.h
```

```
gate_map["h"](0)    # เท่ากับ qc.h(0)
```

```
qubits = [0]
gate_map["h"](*[0])              # แกะ [0] → 0 เท่ากับ qc.h(0)

qubits = [0, 1]
two_q_map["cx"](*[0, 1])       # แกะ [0,1] → 0, 1 เท่ากับ qc.cx(0, 1)
```

ส่วนที่ 1: คำนวณ Statevector (amplitude + probability)

```python
sv = Statevector.from_instruction(
    qc.remove_final_measurements(inplace=False) if qc.num_clbits else qc
)
```

ตรงนี้ Qiskit คำนวณ statevector ให้ทั้งหมด — ได้ array ของ amplitude 8 ตัว (2^3):

```
index 0 → |000⟩ → 0.0000      ← ไม่แสดงเพราะ = 0
index 1 → |001⟩ → 0.0000
index 2 → |010⟩ → 0.6124      ← แสดง
index 3 → |011⟩ → 0.6124      ← แสดง
index 4 → |100⟩ → 0.0000
index 5 → |101⟩ → 0.0000
index 6 → |110⟩ → 0.3536      ← แสดง
index 7 → |111⟩ → 0.3536      ← แสดง
```

ส่วนที่ 2: คำนวณ Bloch coordinates (x, y, z)

```python
# Partial trace: เอา qubit ที่ไม่สนใจออก → ได้ density matrix 2×2
trace_out = [i for i in range(NUM_QUBITS) if i != q]
dm = partial_trace(sv, trace_out)

# แปลง density matrix → พิกัด Bloch
bx, by, bz = bloch_coords(dm.data)
```

โดย function `bloch_coords` คำนวณจาก density matrix ρ ขนาด 2×2:

```
def bloch_coords(dm):
    rho = np.array(dm)
    x = 2 * np.real(rho[0, 1])       # 2 × Re(ρ₀₁)
    y = 2 * np.imag(rho[1, 0])       # 2 × Im(ρ₁₀)
    z = np.real(rho[0, 0] - rho[1, 1]) # ρ₀₀ - ρ₁₁
    return (x, y, z)
```

ยกตัวอย่าง **q1 หลังผ่าน X gate** → สถานะ = |1⟩:

```
        [ 0  0 ]
   ρ =  [ 0  1 ]

   x = 2 × Re(0)   = 0
   y = 2 × Im(0)   = 0
   z = 0 - 1        = -1     ← ตรงกับภาพ: (0, 0, -1)
```

```python
lines = [f"Statevector  (layer {layer}):\n"]
for i, amp in enumerate(sv.data):
    prob = np.abs(amp) ** 2
    if prob > 1e-8:
        basis = f"|{i:0{NUM_QUBITS}b}>"
        lines.append(f"  {basis}  ->  {amp:+.4f}   (P={prob:.4f})")
lines.append("")
lines.append("Bloch coords (x, y, z):")
for qinfo in rec["qubits"]:
    q  = qinfo["qubit"]
    bx, by, bz = qinfo["bloch"]
    r  = np.sqrt(bx**2 + by**2 + bz**2)
    lines.append(f"  q{q}: ({bx:+.4f}, {by:+.4f}, {bz:+.4f})  r={r:.4f}")
```

ตัวเลขทั้งหมดในภาพ ไม่ได้คำนวณเอง

— Qiskit คำนวณให้ผ่าน Statevector.from_instruction() และ partial_trace() แล้วโค้ดเราแค่ดึงค่าออกมาแสดงผล